

Vietnam National University, Ho Chi Minh City
University of Technology
Faculty of Computer Science and Engineering



PF4DAV Assignment

Stroke Prediction Using Tabular Machine Learning

**Course: Programming Foundation for Data Analytics and
Visualization**

Instructor: Dr. Lê Thành Sách

Students:	Nguyễn Lâm Huy	2311188
	Nguyễn Tất Kiên	2311735
	Nguyễn Thành Minh Khôi	2311687

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Duty Roster

No.	Name	StuID	Work	% Progress
1	Nguyễn Tất Kiên	2311735	Image Part Assignment	100%
2	Nguyễn Thành Minh Khôi	2311687	Tabular Part Assignment	100%
3	Nguyễn Lâm Huy	2311188	Text Part Assignment	100%

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1 Introduction

Stroke is a major medical emergency in which early risk identification can support timely screening and preventive care. This project studies whether routinely collected demographic, lifestyle, and clinical attributes can be used to estimate a patient's stroke risk. The task is formulated as a binary tabular classification problem, where the target indicates whether a patient has experienced a stroke.

The project uses the public *Stroke Prediction Dataset*, which contains 5,110 patient records and attributes such as age, hypertension, heart disease, average glucose level, body mass index (BMI), smoking status, work type, and residence type. The dataset presents several practical machine-learning challenges: BMI contains missing values, numerical variables contain skewed distributions and outliers, categorical variables require encoding, and only 249 records belong to the stroke-positive class. Consequently, a model can obtain high accuracy by predicting the majority class while still failing to identify patients at risk.

The objective is therefore not only to maximize overall accuracy, but also to obtain useful recall for the minority stroke class. The workflow includes missing-value analysis, exploratory data analysis, outlier treatment, feature engineering, class balancing, feature reduction, hyperparameter tuning, model comparison, and threshold optimization. Four classifiers are evaluated: Logistic Regression, Random Forest, a Multi-Layer Perceptron (MLP), and XGBoost. An AUC-weighted soft-voting ensemble is also constructed to combine their probability estimates.

The main contributions of the project are as follows:

- A complete preprocessing pipeline for mixed numerical and categorical health data.
- A comparison of K-Nearest Neighbour and Decision Tree approaches for BMI imputation.
- Domain-inspired features describing age, glucose, BMI, cardiovascular conditions, and their interactions.
- Imbalance-aware training using BorderlineSMOTE and the F_2 score, which assigns greater importance to recall.
- A comparison of single models, threshold-tuned models, and a weighted soft-voting ensemble.

The experimental results show that Logistic Regression provides the strongest ranking performance among the individual models, reaching a test ROC AUC of 0.8272. The soft-voting ensemble reaches 0.8000 recall at a threshold of 0.22, demonstrating the trade-off between detecting more stroke cases and producing more false-positive alerts.

2 Dataset and Exploratory Analysis

2.1 Dataset Description

The dataset is obtained from the Kaggle *Stroke Prediction Dataset* published by fedesoriano and loaded in the notebook through `kagglehub`. It contains 5,110 observations and 12 original columns, including the target variable `stroke`. The source is available at <https://www.kaggle.com/datasets/fedesoriano/stroke-prediction-dataset>.

Table 1: Original dataset attributes

Attribute	Type	Description
<code>id</code>	Identifier	Unique patient identifier; removed before modeling.
<code>gender</code>	Categorical	Male, Female, or Other.
<code>age</code>	Numerical	Patient age in years.
<code>hypertension</code>	Binary	Indicates whether the patient has hypertension.
<code>heart_disease</code>	Binary	Indicates whether the patient has heart disease.
<code>ever_married</code>	Categorical	Whether the patient has ever been married.
<code>work_type</code>	Categorical	Employment category, such as private, government, or self-employed.
<code>Residence_type</code>	Categorical	Rural or urban residence.
<code>avg_glucose_level</code>	Numerical	Average blood glucose measurement.
<code>bmi</code>	Numerical	Body mass index.
<code>smoking_status</code>	Categorical	Smoking history; Unknown denotes unavailable information.
<code>stroke</code>	Binary target	1 for stroke and 0 for no stroke.

2.2 Missing-Value Analysis

The initial inspection finds 201 missing values, all in the `bmi` column. Removing these rows would discard approximately 3.9% of the dataset and could remove valuable stroke-positive observations. Two imputation strategies are therefore explored:

- **KNN imputation:** categorical variables are numerically encoded and each missing BMI is estimated from nearby observations. The exploratory comparison uses 15 neighbours.
- **Decision Tree regression:** BMI is predicted from age and gender using a Decision Tree Regressor after scaling the input variables.

Both methods produce plausible BMI values without deleting records. KNN imputation is retained in the final preprocessing pipeline because it can use the broader multivariate neighbourhood. To reduce information leakage, the final imputer is fitted only on the training split and uses five neighbours.

2.3 Numerical Distributions

The three principal continuous variables are age, average glucose level, and BMI. Average glucose level is strongly right-skewed and shows a second concentration at high glucose values.

BMI is also right-skewed, with most observations concentrated around the overweight range. Age covers a broad interval from childhood to 82 years.

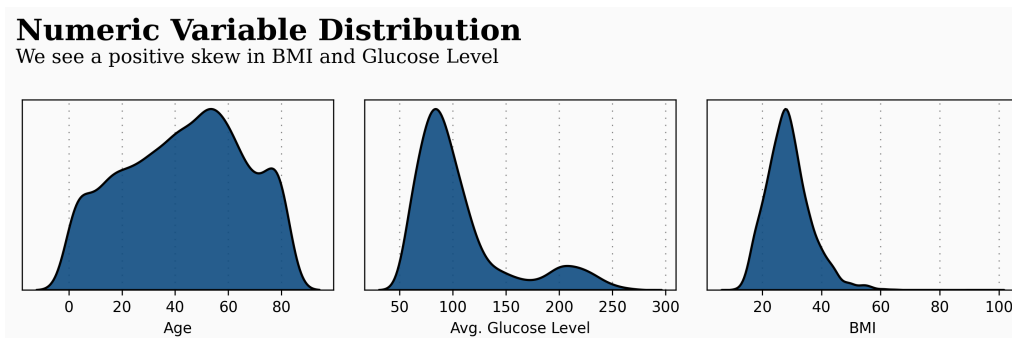


Figure 1: Distributions of age, average glucose level, and BMI

Comparing stroke and non-stroke groups suggests that age is the clearest individual risk indicator. Stroke-positive records are concentrated among older patients, while glucose and BMI show weaker separation. The notebook's cumulative analysis also shows that stroke risk increases with age.

2.4 Class Imbalance

Only 249 of the 5,110 patients have a positive stroke label, corresponding to approximately 4.87% of the dataset. The remaining 4,861 observations belong to the no-stroke class. This imbalance makes accuracy alone unsuitable for model selection because a classifier that predicts only the majority class would already exceed 95% accuracy.



Figure 2: Stroke prevalence is approximately one positive case in every twenty records

The exploratory plots indicate that stroke-positive patients tend to be older and more frequently have hypertension or heart disease. Differences associated with gender, residence type, work type, smoking status, glucose, and BMI are present but overlap substantially between the two classes.

Figure 3 compares the distributions of several attributes between patients with and without stroke. Age provides the clearest separation: stroke-positive records are concentrated among older patients, whereas the no-stroke group is distributed more broadly across younger and middle-aged populations. Hypertension and heart disease are also more common in the stroke group, supporting their use as cardiovascular risk indicators. Average glucose levels tend to be higher and more widely distributed among stroke-positive patients, although the two groups still overlap substantially. In contrast, BMI, gender, smoking status, and work type show weaker visual separation. Private and self-employed patients account for many stroke cases, but these categories are also common in the overall dataset, so their raw counts should not be interpreted as

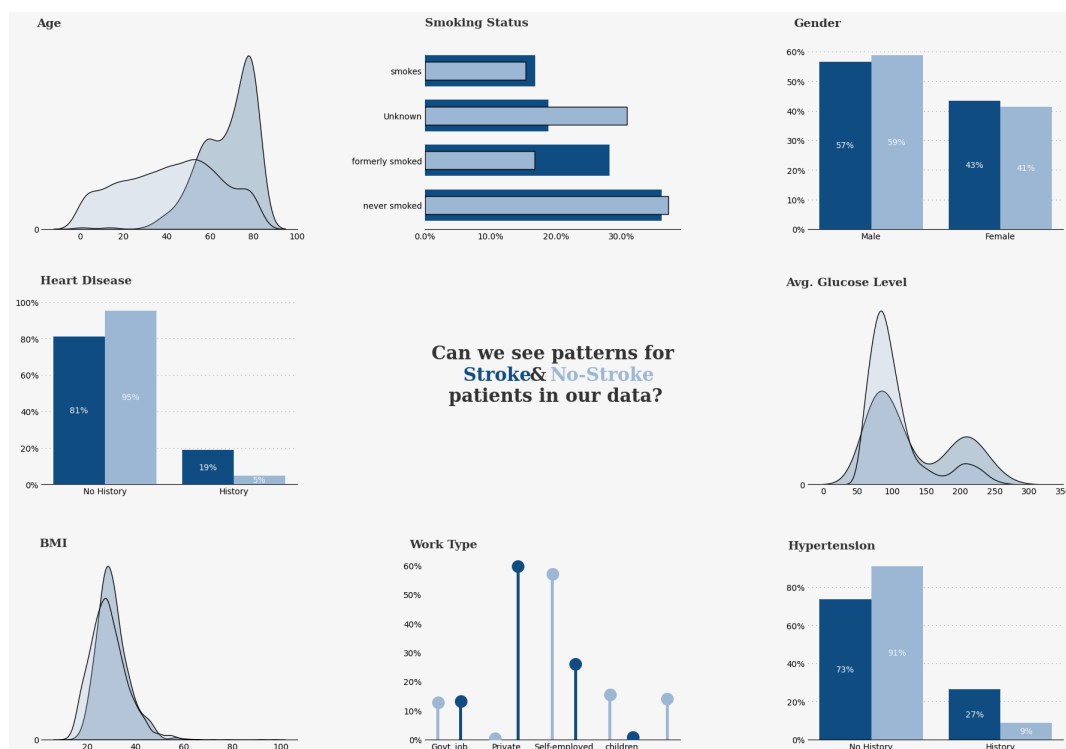


Figure 3: Comparison of selected patient characteristics by stroke label

direct causal effects. Overall, the figure suggests that age and existing cardiovascular conditions provide stronger predictive information than the demographic and lifestyle categories considered individually.

2.5 Outlier Detection and Treatment

Outliers are detected using the interquartile-range rule with fences $Q_1 - 1.5 IQR$ and $Q_3 + 1.5 IQR$. Rows are not removed because the positive class is already scarce. Instead, extreme numerical values are capped to the calculated bounds.

Table 2: Outliers detected before capping

Variable	Lower bound	Upper bound	Outliers	Percentage
Age	-29.00	115.00	0	0.00%
Average glucose level	21.98	169.36	627	12.27%
BMI	9.10	47.50	110	2.24%

Age remains unchanged because all observed ages fall within the IQR bounds. Average glucose is capped from a maximum of 271.74 to 169.36, while BMI is capped from 97.60 to 47.50. Capping preserves every patient record and prevents extreme values from dominating distance-based and gradient-based methods.

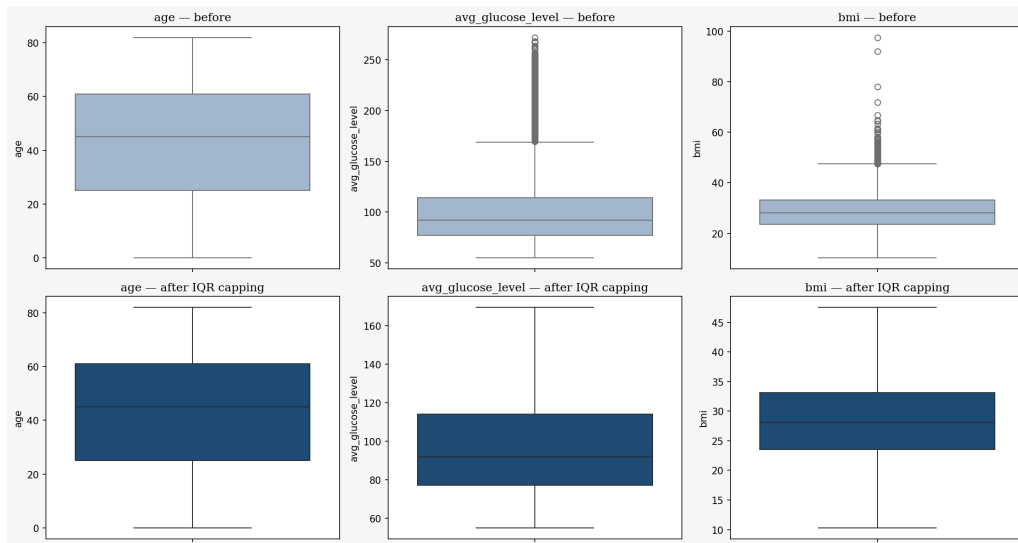


Figure 4: Continuous variables before and after IQR-based capping

3 Proposed Method

3.1 Feature Engineering

The patient identifier is removed because it does not contain clinical information. Since the original variables have weak linear relationships with the target, the notebook creates nonlinear

transformations, binary indicators, ratios, and interaction features. The main engineered groups are:

- **Age features:** squared age, logarithmic age, and young, middle-aged, and older indicators.
- **Glucose features:** squared and logarithmic glucose, together with high-glucose and very-high-glucose indicators.
- **BMI features:** squared and logarithmic BMI and indicators for underweight, overweight, and obesity.
- **Interaction features:** age multiplied by hypertension, heart disease, glucose, BMI, and cardiovascular-risk indicators.
- **Ratio features:** age-to-BMI, age-to-glucose, and BMI-to-glucose ratios.
- **Combined clinical features:** cardiovascular risk and the presence of either hypertension or heart disease.

After feature engineering, each record contains 35 predictors before one-hot encoding. The strongest training-set correlations with stroke are observed for squared age (0.271), age–glucose interaction (0.251), age (0.242), the older-age indicator (0.220), and age–BMI interaction (0.206). These values confirm that age-related transformations contain the clearest predictive signal.

3.2 Data Splitting and Preprocessing

The data is divided using stratified sampling so that each split preserves the original class ratio:

- Training set: 3,066 records (60%).
- Validation set: 1,022 records (20%).
- Test set: 1,022 records (20%).

Numerical features are processed with a five-neighbour KNN imputer followed by standardization. Categorical features are transformed using one-hot encoding with support for unseen categories. The preprocessing transformer is fitted on the training data and then applied unchanged to validation and test data.

3.3 Imbalance Handling

BorderlineSMOTE is applied only to the processed training set. Unlike ordinary random over-sampling, BorderlineSMOTE focuses on minority examples near the decision boundary, where classification is more difficult. After resampling, the training matrix contains 5,834 observations and 46 encoded features with balanced target classes. Validation and test sets retain their original distributions to provide realistic evaluation.

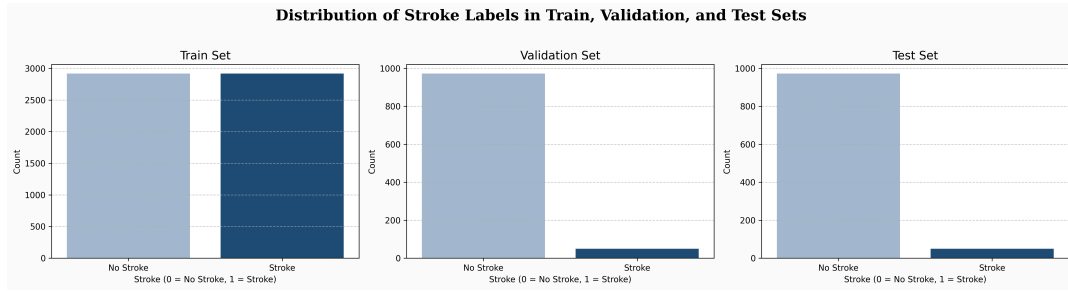


Figure 5: Class distributions in the resampled training set and untouched validation and test sets

3.4 Feature Reduction

Two feature-reduction strategies are used according to model type:

- Random Forest and XGBoost use the 25 highest-ranked features selected by mutual information.
- Logistic Regression and the MLP use Principal Component Analysis (PCA). PCA reduces the 46 encoded features to 11 components while retaining 95.07% of the variance.

3.5 Model Training and Hyperparameter Tuning

The project evaluates four complementary classifiers. Logistic Regression provides a linear and interpretable baseline. Random Forest combines multiple decision trees through bagging. XGBoost uses gradient-boosted trees, while the MLP learns nonlinear relationships through fully connected neural-network layers.

Randomized search with five-fold stratified cross-validation is used for Logistic Regression, Random Forest, and XGBoost. Hyperparameters are selected using the F_2 score:

$$F_2 = \frac{(1 + 2^2) \text{Precision Recall}}{2^2 \text{Precision} + \text{Recall}}. \quad (1)$$

This metric weights recall four times more heavily than precision, reflecting that a missed high-risk patient is more costly than an additional screening alert.

Table 3: Selected model configurations

Model	Configuration
Logistic Regression	$C = 0.0178$, L2 penalty, liblinear solver, balanced class weights; best cross-validation $F_2 = 0.8672$.
Random Forest	370 trees, depth 16, log2 feature sampling, minimum leaf size 6, minimum split size 8; best cross-validation $F_2 = 0.9646$.
XGBoost	516 trees, depth 6, learning rate 0.1926, subsample 0.7124, column sample 0.8083, and regularization; best cross-validation $F_2 = 0.9655$.
MLP	Two hidden layers with 128 and 64 neurons, ReLU activation, Adam optimization, early stopping, and a maximum of 500 iterations.

3.6 Soft-Voting Ensemble

The final ensemble combines predicted probabilities from all four classifiers. Each model receives a weight derived from its validation ROC AUC:

$$w_m = \max(AUC_m - 0.5, 0)^2. \quad (2)$$

The resulting weights are 0.118 for Logistic Regression, 0.101 for Random Forest, 0.067 for XGBoost, and 0.031 for the MLP. The classification threshold is selected on the validation set to maximize F_2 , producing a final threshold of 0.22.

4 Experimental Results

4.1 Evaluation Metrics

The models are evaluated using ROC AUC, precision, recall, F_1 , F_2 , and accuracy. ROC AUC measures ranking quality across thresholds, while precision and recall focus on the positive stroke class. Because the dataset is strongly imbalanced, recall and F_2 are particularly important. Accuracy is reported for completeness but is not used as the main indicator of clinical usefulness.

4.2 Results at the Default Threshold

Table 4 reports performance at the default probability threshold of 0.5. Logistic Regression achieves the highest test ROC AUC and the highest minority-class recall. XGBoost obtains the highest accuracy, but its recall of 0.12 shows that it misses most stroke-positive patients.

Table 4: Test performance at the default threshold of 0.5

Model	ROC AUC	Precision	Recall	F_1	Accuracy
Logistic Regression	0.8272	0.16	0.78	0.26	0.7828
Random Forest	0.8133	0.29	0.18	0.22	0.9384
XGBoost	0.7666	0.27	0.12	0.17	0.9413
MLP Classifier	0.7133	0.13	0.12	0.12	0.9178

The result illustrates why overall accuracy can be misleading. Random Forest, XGBoost, and the MLP achieve accuracy above 0.91 primarily by predicting the dominant no-stroke class. Logistic Regression accepts lower accuracy in exchange for identifying 78% of the stroke cases.

4.3 Threshold Optimization

The default threshold is not necessarily suitable for an imbalanced medical-screening problem. Therefore, each model's threshold is selected on the validation set to maximize F_2 . The tuned test results are shown in Table 5.

Logistic Regression remains the best threshold-tuned individual model with $F_2 = 0.4388$. Random Forest and XGBoost require very low thresholds to recover more positive cases, which substantially reduces precision and accuracy. These results indicate that their probability outputs are not well aligned with the rare positive class even though their majority-class accuracy is high.

Table 5: Test results after validation-based threshold optimization

Model	Threshold	ROC AUC	PR AUC	Precision	Recall	F_1	F_2	Accuracy
Logistic Regression	0.617	0.8272	0.2308	0.1875	0.6600	0.2920	0.4388	0.8434
Random Forest	0.140	0.8133	0.1926	0.1259	0.7400	0.2151	0.3745	0.7358
XGBoost	0.010	0.7666	0.1641	0.1090	0.6800	0.1878	0.3320	0.7123
MLP Classifier	0.014	0.7133	0.1220	0.1207	0.4200	0.1875	0.2807	0.8219

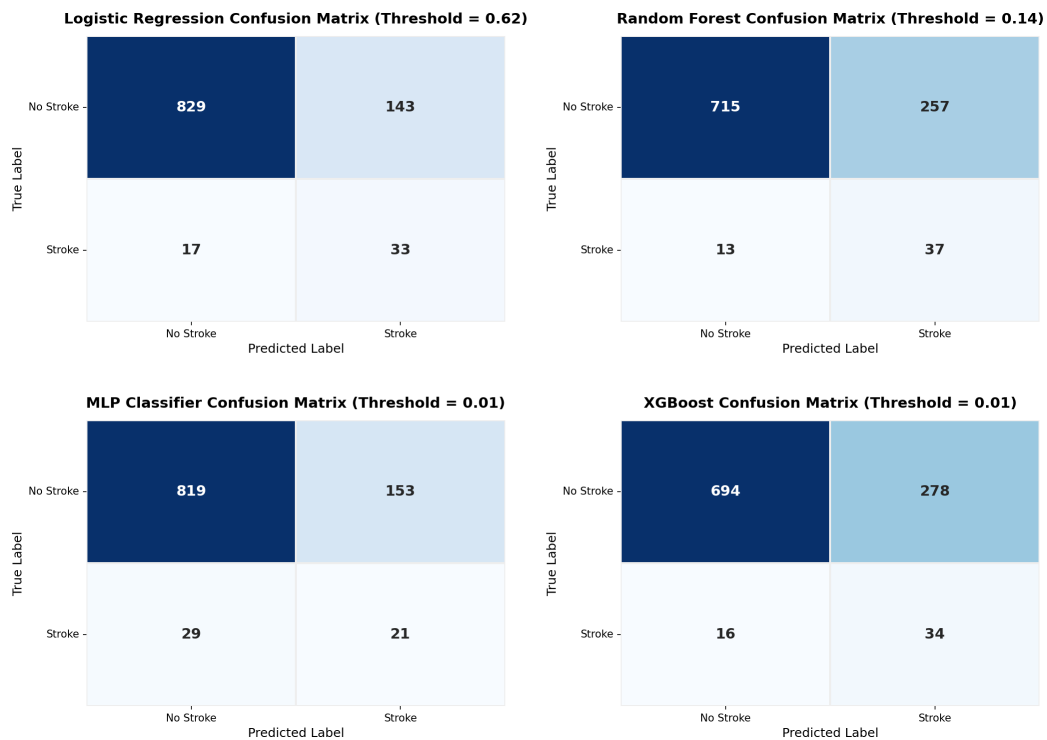


Figure 6: Confusion matrices using validation-tuned thresholds

4.4 Ensemble Performance

The AUC-weighted soft-voting ensemble is evaluated at its validation-selected threshold of 0.22. Its test results are presented in Table 6.

Table 6: AUC-weighted soft-voting ensemble performance

Metric	Score
Threshold	0.22
ROC AUC	0.8234
PR AUC	0.2042
Precision	0.1404
Recall	0.8000
F_1	0.2388
F_2	0.4124
Accuracy	0.7505

The ensemble has slightly lower ROC AUC than Logistic Regression but reaches the highest recall among the final threshold-selected systems. It identifies 80% of stroke-positive patients, although precision falls to 14.04%. This operating point is suitable only when the system is treated as an initial screening tool and false-positive predictions can be followed by professional clinical assessment.

4.5 Discussion

The experiments reveal four main findings. First, age-related features contain most of the useful predictive signal. Second, class imbalance has a much larger effect on practical performance than headline accuracy suggests. Third, the comparatively simple Logistic Regression model generalizes better than the more complex MLP and boosted-tree model in terms of ROC AUC. Finally, threshold selection changes the interpretation of each classifier and must be chosen according to the cost of false negatives and false positives.

The strong cross-validation F_2 values obtained by Random Forest and XGBoost on the BorderlineSMOTE-resampled data do not transfer fully to the naturally imbalanced test set. This gap suggests that synthetic minority examples make the training task easier than real-world evaluation. It also highlights the importance of keeping validation and test distributions untouched.

5 Limitations and Future Work

5.1 Limitations

- The dataset contains only 249 positive cases, limiting the diversity of stroke patterns available for learning.
- BorderlineSMOTE creates synthetic samples but cannot introduce new clinical information.
- IQR capping is performed before the final split in the notebook; a production pipeline should estimate all clipping bounds exclusively from training data.

- Several clinically important variables are unavailable, including blood pressure measurements, cholesterol, medication, family history, and previous transient ischemic attacks.
- The models are evaluated on a single public dataset and have not been externally validated on another population.
- Probability calibration is not assessed, so predicted probabilities should not be interpreted directly as clinical risk percentages.

5.2 Future Work

Future experiments should place imputation, clipping, encoding, sampling, feature selection, and classification inside a leakage-safe cross-validation pipeline. Nested cross-validation would provide a less biased estimate of hyperparameter and threshold selection. Cost-sensitive learning and focal-loss approaches could be compared with synthetic oversampling. CatBoost is also a promising alternative because it can handle categorical variables directly. Finally, probability calibration, interpretability methods such as SHAP, and external clinical validation are necessary before the model could support real screening decisions.

6 Conclusion

This project develops an end-to-end tabular machine-learning workflow for stroke prediction. The workflow addresses missing BMI values, skewed numerical variables, outliers, mixed feature types, weak raw correlations, and severe class imbalance. Feature engineering and imbalance-aware evaluation improve the model's ability to detect the rare stroke class.

Logistic Regression is the strongest individual model, achieving a test ROC AUC of 0.8272 and the best threshold-tuned F_2 score of 0.4388. The AUC-weighted ensemble provides the highest final recall of 0.8000 but has low precision. The results demonstrate that stroke prediction in this dataset is primarily a ranking and screening problem rather than a conventional accuracy-maximization task. The system may be useful as a research prototype, but further validation and clinically richer data are required before deployment.